## Jargon Buster

**ISO9660 (CDFS):** The most commonly used file system for CD and DVD is ISO9600. It has certain limitations that led to the development of new file systems. The limitations of this file system were short file names and eight-level-deep file structures, which means that you can create only seven sub-folders under a root. The newer file systems resolve this issue.

We cannot totally do away with CDFS, since you'll need it to view the contents of a CD in DOS-mode or older Mac, Sun or Linux systems. However, there are extensions to ISO to deal with its limitations. These are Rock Ridge and Joliet.

**Rock Ridge:** The Rock Ridge Interchange Protocol (RRIP) assigns an extension to the ISO9660 standard for CD-ROM that not only lifts the ISO9660 restrictions, but also allows *nix-style symbolic links and special files to be stored on a CD. The volume descriptors are not affected by this extension, so it is cross-platform.

**Joliet:** Developed by Microsoft, this is an extension to ISO9660 that allows Unicode characters in file names. It also allows long file names just like Rock Ridge, but the limitation is that only Linux and FreeBSD among the POSIX systems can recognise Joliet CDs.

**Hybrid CD:** In order to eliminate cross-platform issues when dealing with CD-ROM/RW content, the Hybrid class was designed. It is a mixture of three file systems on a single disc—ISO9660/RR, Joliet and HFS. Such CDs are accessible under DOS, *nix, Macintosh, and Windows 9x/NT. It also allows more than one session on a disc, which means that a recordable CD can be used more than once—provided there's remaining space, and that the disc wasn't closed in the previous burn session.

**HFS:** HFS is a file system developed by Apple computers for computers running Macintosh operating systems. It was originally designed for floppy disks and hard disks, but it can now also be found on CD-ROMs. It permits file names up to 31 characters, and supports metadata (information about data content) on a CD. It is a well-documented file system, and so many workarounds are available to access HFS-based CDs on other modern operating systems.

**UDF:** Universal Disk Format (UDF) is a relatively newer file system for CD and DVD that is developed and managed by the Optical Storage Technology Association (OSTA), aimed at creating a platform-independent media format.

It is based on the ISO 13346 (EMCA-167) standard, and it is considered the successor to the ISO 9660 (CDFS) format. UDF supports advanced features along with Long and Unicode file names, deep directory trees, large (64-bit) file sizes, access control list, and named streams.

However, Windows XP's inbuilt CD-burning feature does not support some UDF features such as named streams and access control list features.

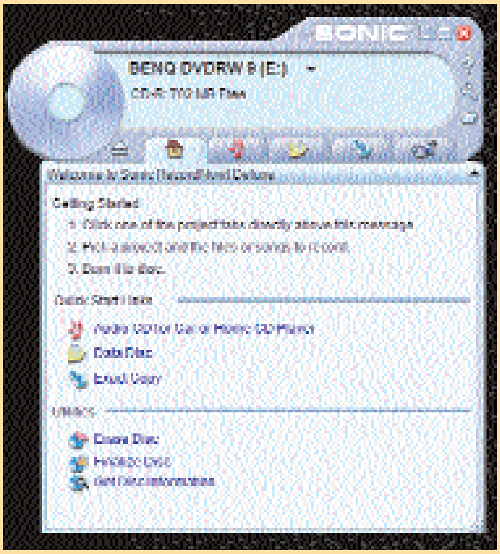
UDF is required by DVDs to contain MPEG audio/video streams. It is also used by CD-RW in the packet writing process, which ensures efficiency in terms of disk space required and time.

**TAO:** Track-At-Once (TAO) is a method of disc writing by which the laser writes data on a disc track-by-track. The laser of the device goes on and off between these tracks. If an audio disc is written using TAO, it will introduce rather long gaps (3 or 4 seconds) between audio tracks, which cannot be reduced by the burning software.

**DAO:** In Disc-At-One, one or more tracks are written continuously one after the other; the laser remains on till the process is complete. This method is ideal for burning audio discs, and most CD recording software use it by default when a user wants to create an audio CD. It is also an ideal option for burning disc images such as *.ISO.

**CRC (Cyclic Redundancy Check):** A CRC error happens when a data check at the destination does not match the original data. This error is commonly seen when writing CDs and DVDs. You'll also see it in downloaded files and corrupt zipped files.

When data is transferred between a source and a destination, a CRC value is given to a block of data. If something goes wrong with the transfer, the CRC sent at the source will not match with that at the destination. When this happens, the OS returns a CRC error.



The interface of Sonic RecordNow! is so compact that it looks more like an MP3 playback software like Jet Audio rather than a burning software

various burn tasks into audio, video, data, etc. You can choose to drag and drop the files you want to burn using Windows Explorer, since the file browser is as compact as the interface. The interface is intuitive - drag MP3 tracks onto the disc symbol on the interface, and it will automatically open the MP3 CD creator. You can also choose to change the volume levels of the sound tracks you add.

### Performance

RecordNow!, too, was very low on resources right from the opening of the application to the end of the burn process. The time taken to record disc images and data were average. The readings we obtained were just the values we expected from an application with such a modest interface.

**Site: www.roxio.com/en/products/index.jhtml**

### So What Software To Buy?

Deciding the winner meant choosing between Nero 7 Premium and Sonic/Roxio Easy Media Creator 8. The other products in close competition were NTI CD&DVD Maker 7 and Cyber-Link's Power2Go. The latter is a better choice for its simplicity,